

Support vector machine

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Large margin classifier

Part 1

Logistic regression

□ Hypothesis:

- $h_{\theta}(x) = g(\theta^T x)$
- $h_{\theta}(x) = P(y = 1|x; \theta)$

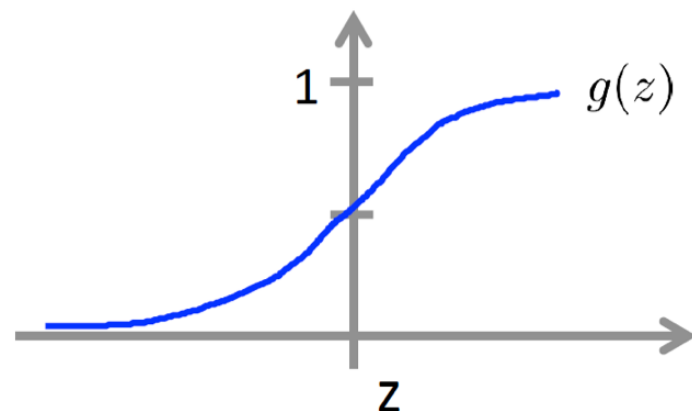
$$g(z) = \frac{1}{1 + e^{-z}}$$

□ $\theta^T x \gg 0$

- Possibility $y = 1$ is higher

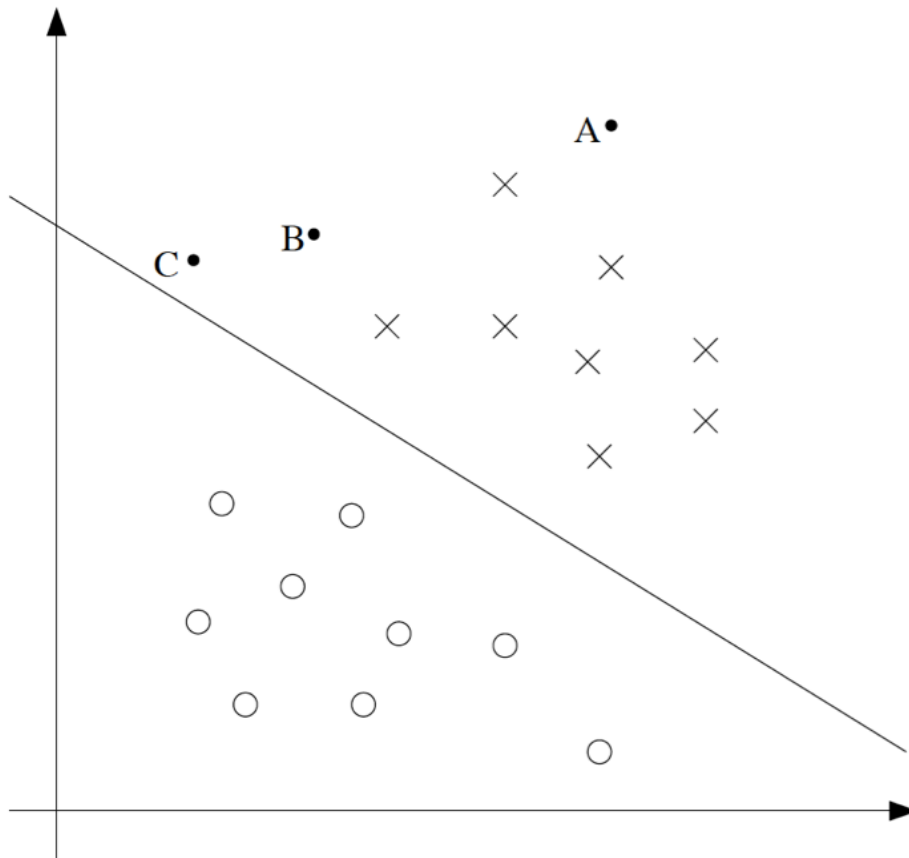
□ $\theta^T x \ll 0$

- Possibility $y = 0$ is higher



Logistic regression

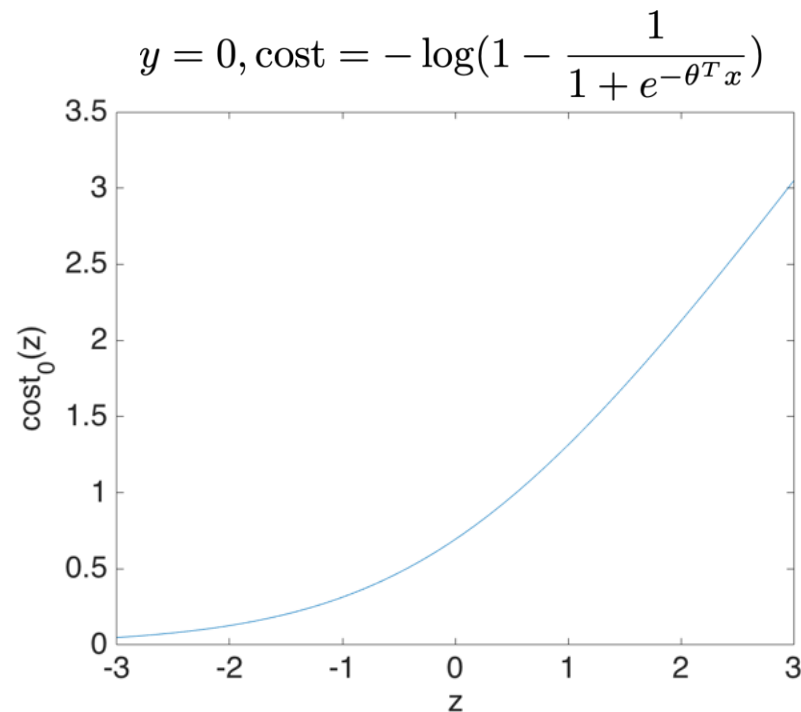
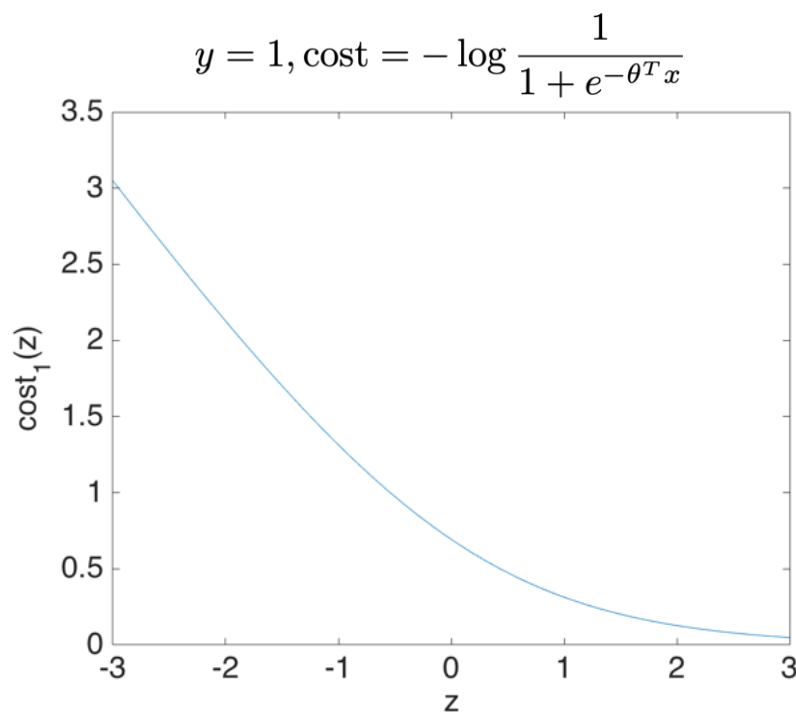
□ Margin



Logistic regression

□ Cost function for a sample

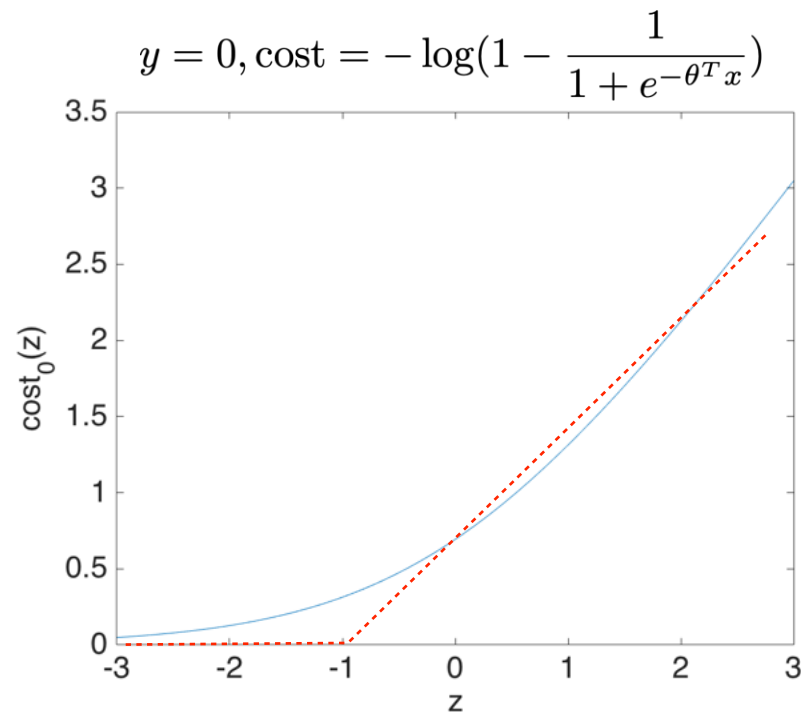
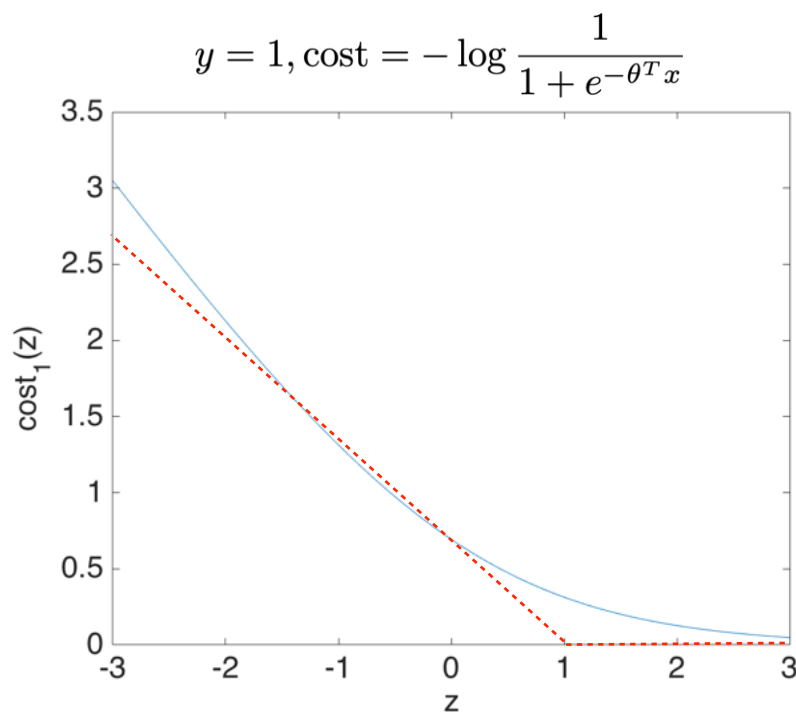
$$-y \log \frac{1}{1 + e^{-\theta^T x}} - (1 - y) \log \left(1 - \frac{1}{1 + e^{-\theta^T x}}\right)$$



Logistic regression

□ Cost function for a sample

$$-y \log \frac{1}{1 + e^{-\theta^T x}} - (1 - y) \log \left(1 - \frac{1}{1 + e^{-\theta^T x}}\right)$$



Support vector machine

□ Cost function

$$J(\theta) = C \sum_{i=1}^m \left[y^{(i)} \text{cost}_1(\theta^T x^{(i)}) + (1 - y^{(i)}) \text{cost}_0(\theta^T x^{(i)}) \right] + \frac{1}{2} \sum_{j=1}^n \theta_j^2$$

□ In comparison with logistic regression

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \left[y^{(i)} (-\log h_{\theta}(x^{(i)})) + (1 - y^{(i)}) (-\log(1 - h_{\theta}(x^{(i)}))) \right] + \frac{\lambda}{2m} \sum_{j=1}^n \theta_j^2$$

SVM model

□ Hypothesis

- $y = 1$ if $\theta^T x \geq 1$
- $y = 0$ if $\theta^T x \leq -1$

□ Cost function

$$J(\theta) = C \sum_{i=1}^m \left[y^{(i)} \text{cost}_1(\theta^T x^{(i)}) + (1 - y^{(i)}) \text{cost}_0(\theta^T x^{(i)}) \right] + \frac{1}{2} \sum_{j=1}^n \theta_j^2$$

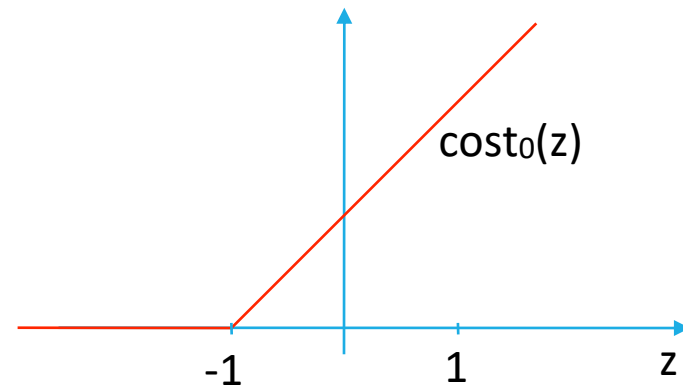
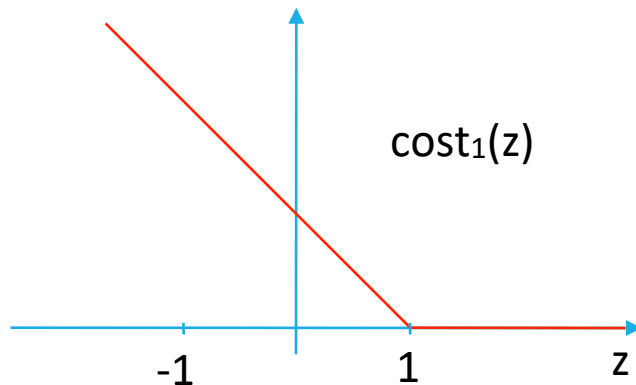
□ Learning:

$$\min_{\theta} J(\theta)$$

Learning

□ Training:

$$\min_{\theta} C \sum_{i=1}^m \left[y^{(i)} \text{cost}_1(\theta^T x^{(i)}) + (1 - y^{(i)}) \text{cost}_0(\theta^T x^{(i)}) \right] + \frac{1}{2} \sum_{j=1}^n \theta_j^2$$



- If $y = 1$, we need $\theta^T x \geq 1$
- If $y = 0$, we need $\theta^T x \leq -1$

Learing

□ Goal

$$\min_{\theta} C \sum_{i=1}^m \left[y^{(i)} \text{cost}_1(\theta^T x^{(i)}) + (1 - y^{(i)}) \text{cost}_0(\theta^T x^{(i)}) \right] + \frac{1}{2} \sum_{j=1}^n \theta_j^2$$

- If $y = 1$, we need $\theta^T x \geq 1$
- If $y = 0$, we need $\theta^T x \leq -1$

$$\rightarrow \min_{\theta} \frac{1}{2} \sum_{j=1}^n \theta_j^2$$

$$\text{s.t. } \theta^T x^{(i)} \geq 1 \quad \text{if } y^{(i)} = 1$$

$$\theta^T x^{(i)} \leq -1 \quad \text{if } y^{(i)} = 0$$

Why large margin

- Dot product

$$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$u^T v = u_1 v_1 + u_2 v_2$$

- Magnitude of vector u : $\|u\| = \sqrt{u_1^2 + u_2^2}$

- Projection of v on u : p

- Then: $u^T v = p \|u\|$

Why large margin



$$\min_{\theta} \frac{1}{2} \sum_{j=1}^n \theta_j^2$$

$$\text{s.t. } \theta^T x^{(i)} \geq 1 \quad \text{if } y^{(i)} = 1$$

$$\theta^T x^{(i)} \leq -1 \quad \text{if } y^{(i)} = 0$$



Dot product:

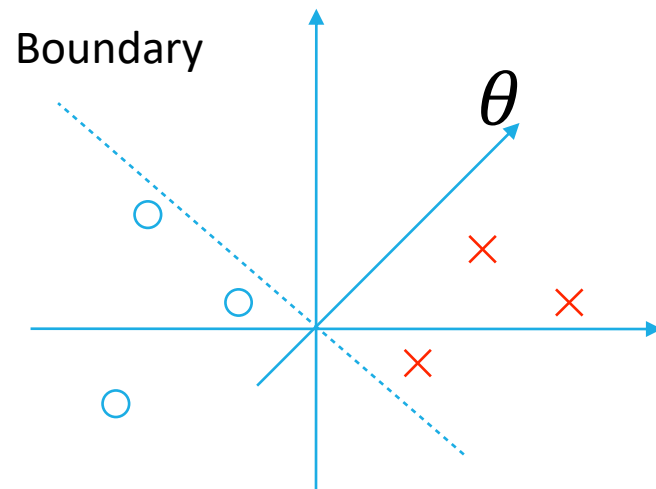
$$\begin{aligned} \theta^T x^{(i)} &= p^{(i)} \|\theta\| \\ &= \theta_1 x_1^{(i)} + \theta_2 x_2^{(i)} \end{aligned}$$

Why large margin

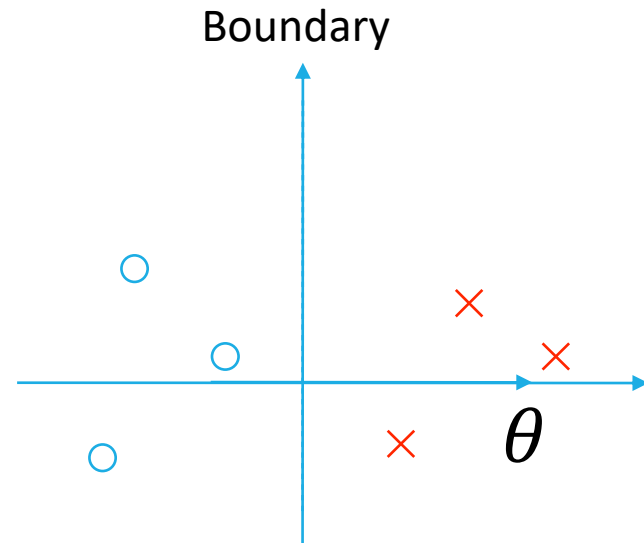
- $$\min_{\theta} \frac{1}{2} \sum_{j=1}^n \theta_j^2$$

s.t. $p^{(i)} \|\theta\| \geq 1$ if $y^{(i)} = 1$
 $p^{(i)} \|\theta\| \leq -1$ if $y^{(i)} = 0$
- Given $p^{(i)}$ to be length of projection of $x^{(i)}$ on θ
- $p^{(i)}$ is distance between sample and boundary
- Do $p^{(i)} \|\theta\| \geq 1$
 - $p^{(i)}$ is smaller, length of θ is larger
 - $p^{(i)}$ is large, length of θ can be small

Why large margin



Narrow margin



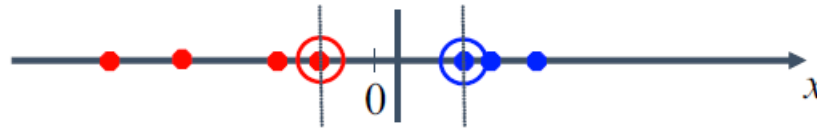
Large margin

Feature mapping with Kernel

Part 2

Non-linear classification

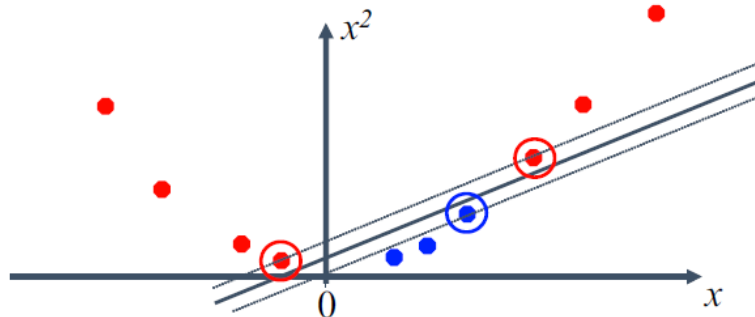
- Linear classifier works well on linearly separated data



- Some data are not linearly separated

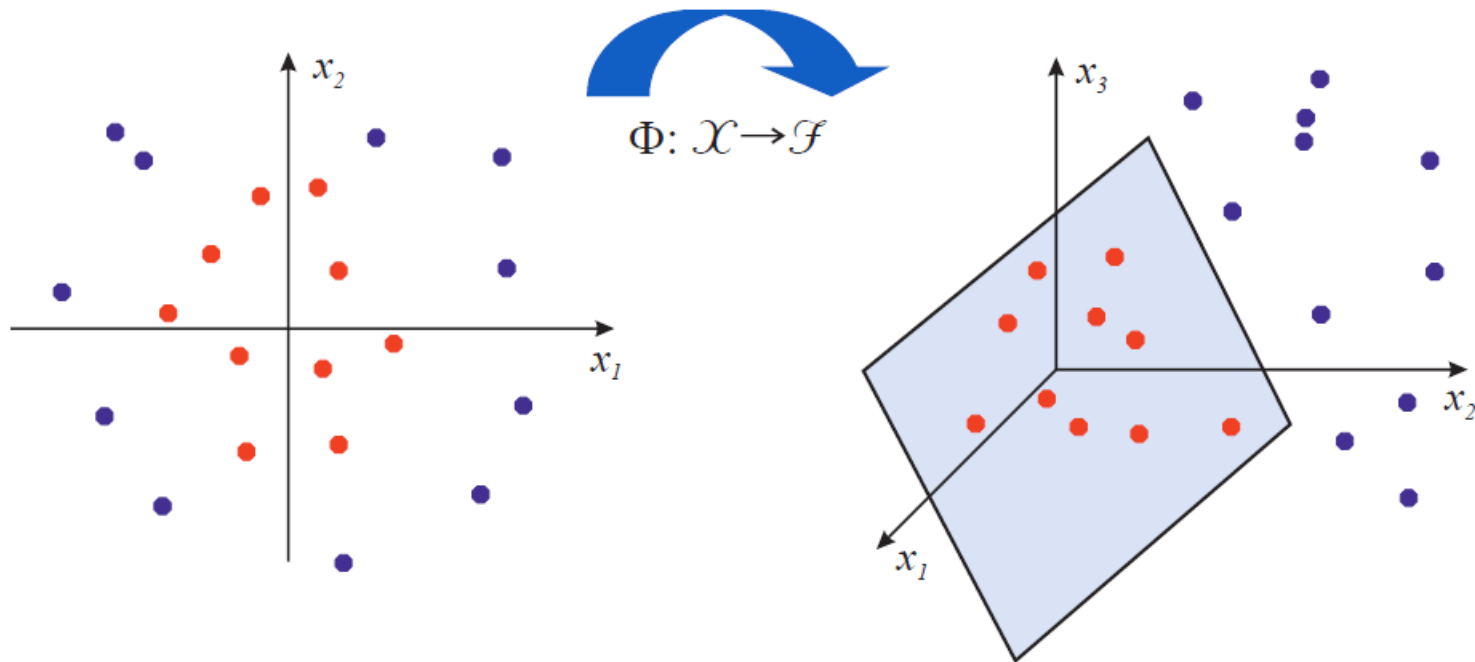


- Solution: map data into higher dimension spaces



Non-linear classification

- Solution: map data into higher dimension spaces



How is data mapping is done with SVM?

Learning algorithm

$$\begin{aligned} \min_{\theta} \quad & \frac{1}{2} \sum_{j=1}^n \theta_j^2 \\ \text{s.t.} \quad & \theta^T x^{(i)} \geq 1 \quad \text{if } y^{(i)} = 1 \\ & \theta^T x^{(i)} \leq -1 \quad \text{if } y^{(i)} = 0 \end{aligned}$$

Given $y = -1$ for negative class

$$\begin{aligned} \Rightarrow \min_{\theta} \quad & \frac{1}{2} \sum_{j=1}^n \theta_j^2 \\ \text{s.t.} \quad & y^{(i)} (\theta^T x^{(i)}) \geq 1, i = 1, 2, \dots, m \end{aligned}$$

Learning algorithm

□ Given $g_i(\theta) = -y^{(i)}(\theta^T x^{(i)}) + 1$

$$\Rightarrow \min_{\theta} \frac{1}{2} \sum_{j=1}^n \theta_j^2$$

$$\text{s.t. } g_i(\theta) = -y^{(i)}(\theta^T x^{(i)}) + 1 \leq 0$$

Switch to Lagrange problem:

$$\mathcal{L}(\theta, \alpha) = \frac{1}{2} \sum_{j=1}^n \theta_j^2 - \sum_{i=1}^m \alpha_i [y^{(i)}(\theta^T x^{(i)}) - 1]$$

Goal: $\min_{\theta, \alpha} \mathcal{L}(\theta, \alpha)$

Learning algorithm

$$\mathcal{L}(\theta, \alpha) = \frac{1}{2} \sum_{j=1}^n \theta_j^2 - \sum_{i=1}^m \alpha_i [y^{(i)} (\theta^T x^{(i)}) - 1]$$

Goal: $\min_{\theta, \alpha} \mathcal{L}(\theta, \alpha)$

Conditions Karush-Kuhn-Tucker (KKT)

Suppose θ^*, α^* to be solution

- (1) $\frac{\partial}{\partial \theta_j} \mathcal{L}(\theta^*, \alpha^*) = 0, j = 1, \dots, n$
- (2) $\alpha_i^* g_i(\theta^*) = 0, i = 1, \dots, m$
- (3) $g_i(\theta^*) \leq 0, i = 1, \dots, m$
- (4) $\alpha_i^* \geq 0, i = 1, \dots, m$

Learning algorithm

$$\mathcal{L}(\theta, \alpha) = \frac{1}{2} \sum_{j=1}^n \theta_j^2 - \sum_{i=1}^m \alpha_i [y^{(i)} (\theta^T x^{(i)}) - 1]$$

Goal: $\min_{\theta, \alpha} \mathcal{L}(\theta, \alpha)$

Solution for theta:

$$(1) \quad \nabla_{\theta} \mathcal{L}(\theta, \alpha) = \theta - \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)} = 0$$

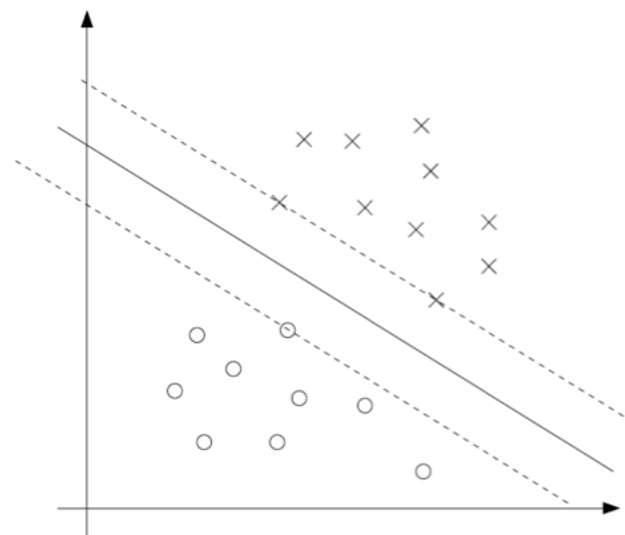
$$\Rightarrow \theta = \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)}$$

Support vector

Conditions Karush-Kuhn-Tucker (KKT)

Suppose θ^*, α^* to be solution

- (1) $\frac{\partial}{\partial \theta_j} \mathcal{L}(\theta^*, \alpha^*) = 0, j = 0, \dots, n$
- (2) $\alpha_i^* g_i(\theta^*) = 0, i = 1, \dots, m$
- (3) $g_i(\theta^*) \leq 0, i = 1, \dots, m$
- (4) $\alpha_i^* \geq 0, i = 1, \dots, m$



From (2), (3) and (4):

if $g_i(\theta^*) < 0$, then $\alpha_i^* = 0, i = 1, \dots, m$

Samples outside margins: $\alpha_i^* = 0$

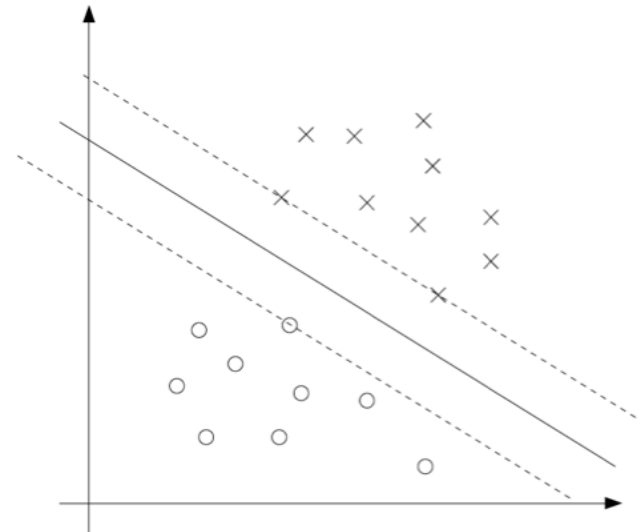
Support vector

Samples outside margins: $\alpha_i^* = 0$

$$\Rightarrow \theta = \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)}$$

Predict:

$$\begin{aligned} \theta^T x &= \left(\sum_{i=1}^m \alpha_i y^{(i)} x^{(i)} \right)^T x \\ &= \sum_{i=1}^m \alpha_i y^{(i)} \langle x^{(i)}, x \rangle \end{aligned}$$



Only calculated with data points on margins $\Rightarrow$ support vector

Feature mapping

- Given a function

$$\phi(x) = [x \quad x^2 \quad x^3]^T$$

Predict:

$$\begin{aligned}\theta^T x &= \left(\sum_{i=1}^m \alpha_i y^{(i)} x^{(i)} \right)^T x \\ &= \sum_{i=1}^m \alpha_i y^{(i)} \langle x^{(i)}, x \rangle\end{aligned}$$

Replace $\langle x^{(i)}, x \rangle$ by $\langle \phi(x^{(i)}), \phi(x) \rangle$

Kernel

□ Given

$$K(x^{(i)}, x) = \langle \phi(x^{(i)}), \phi(x) \rangle$$

Predict:

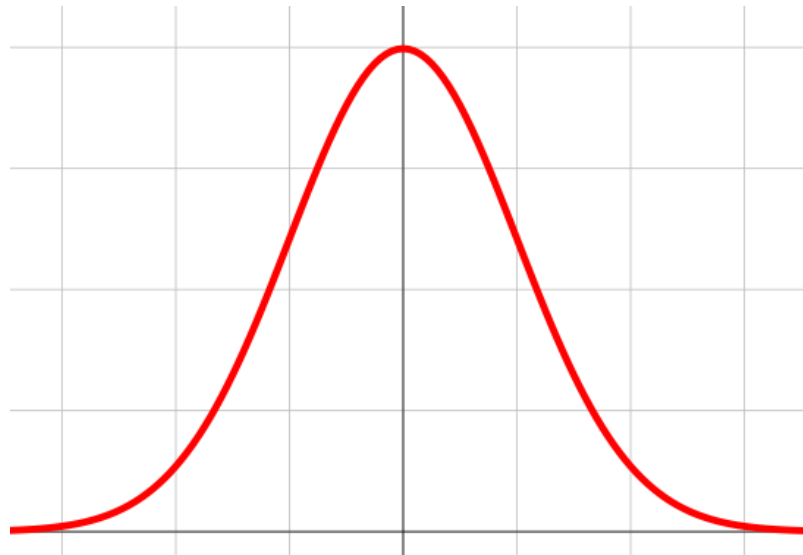
$$\begin{aligned}\theta^T x &= \left(\sum_{i=1}^m \alpha_i y^{(i)} x^{(i)} \right)^T x \\ &= \sum_{i=1}^m \alpha_i y^{(i)} \langle x^{(i)}, x \rangle\end{aligned}$$

Replace $\langle x^{(i)}, x \rangle$ by $K(x^{(i)}, x)$

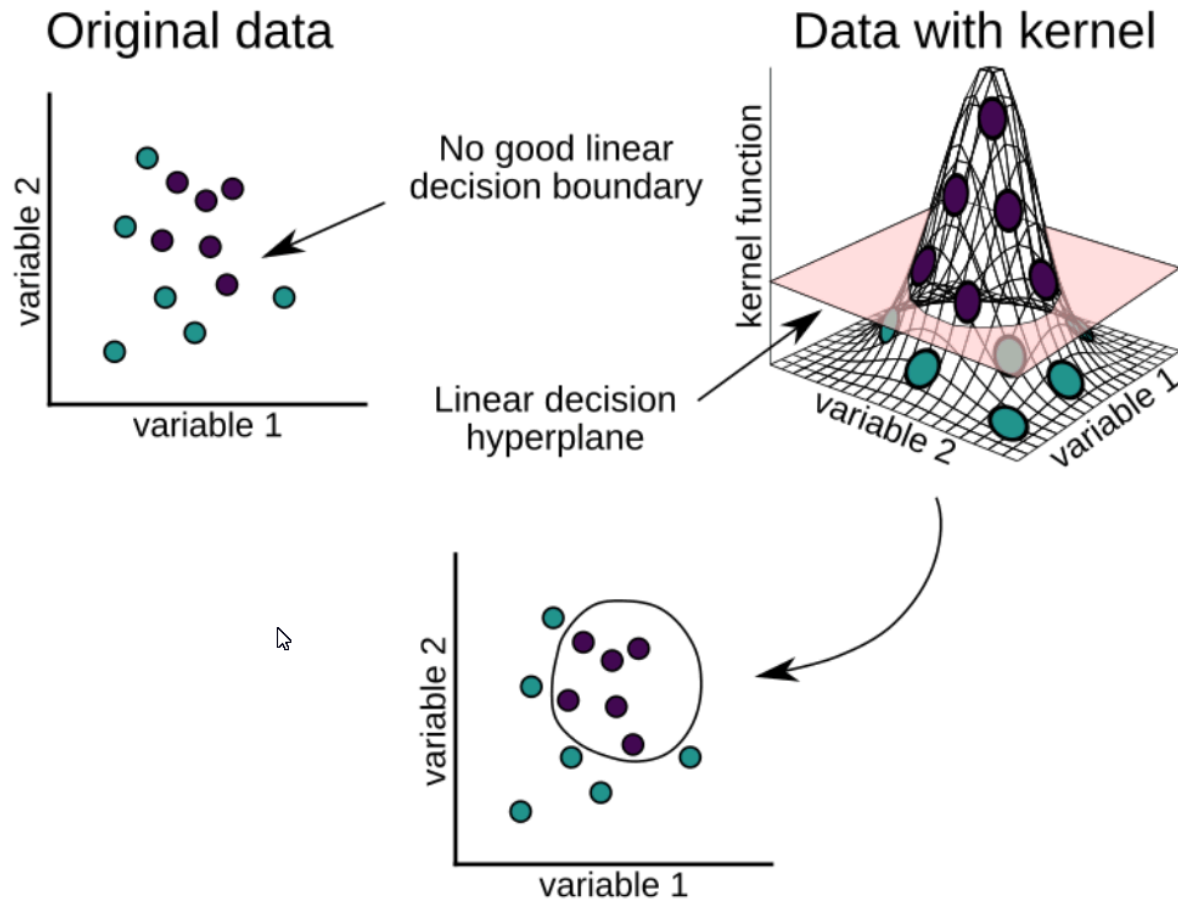
Kernel

□ Gaussian kernel

$$K(x, z) = \exp \left(-\frac{\|x - z\|^2}{2\sigma^2} \right)$$



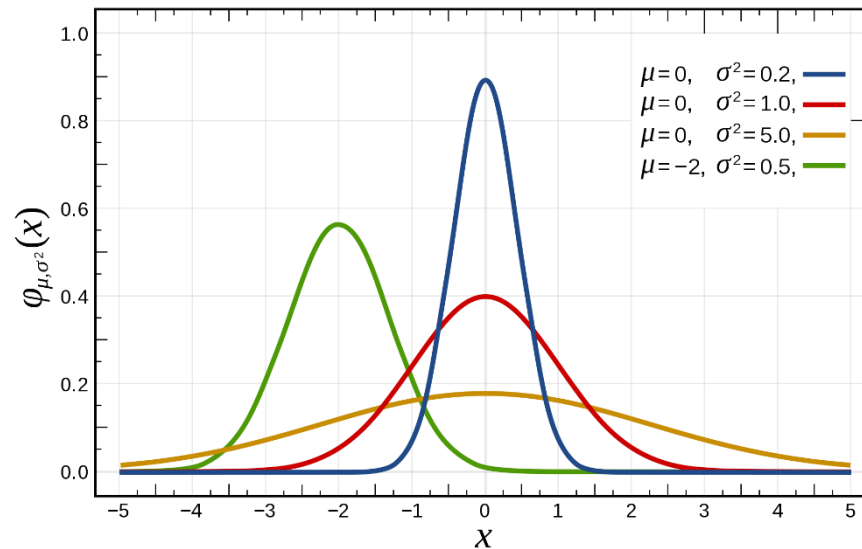
Kernel



Kernel

□ Gaussian (RBF) kernel

$$K(x, z) = \exp\left(-\frac{\|x - z\|^2}{2\sigma^2}\right)$$



- Larger σ , smaller magnitude

Kernel

- Linear kernel

$$K(x, z) = x^T z$$

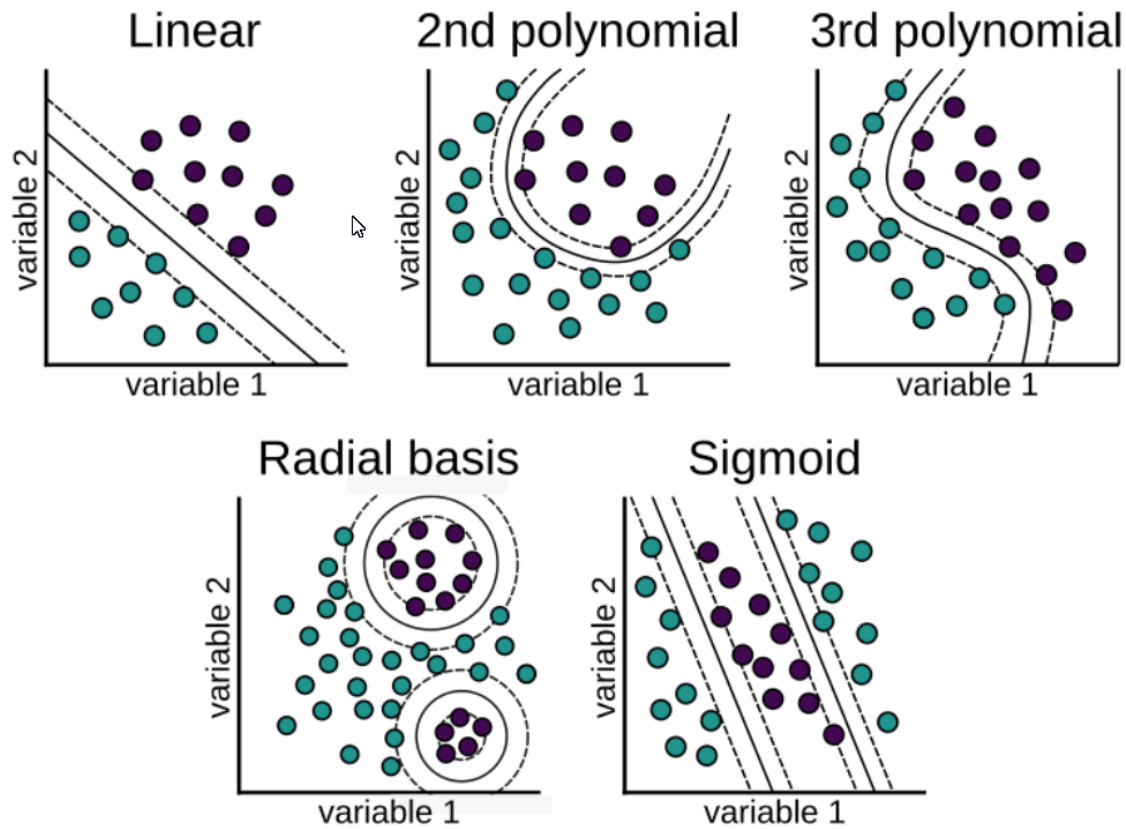
- Polynomial kernel

$$K(x, z) = (x^T z + c)^d$$

- Sigmoid kernel

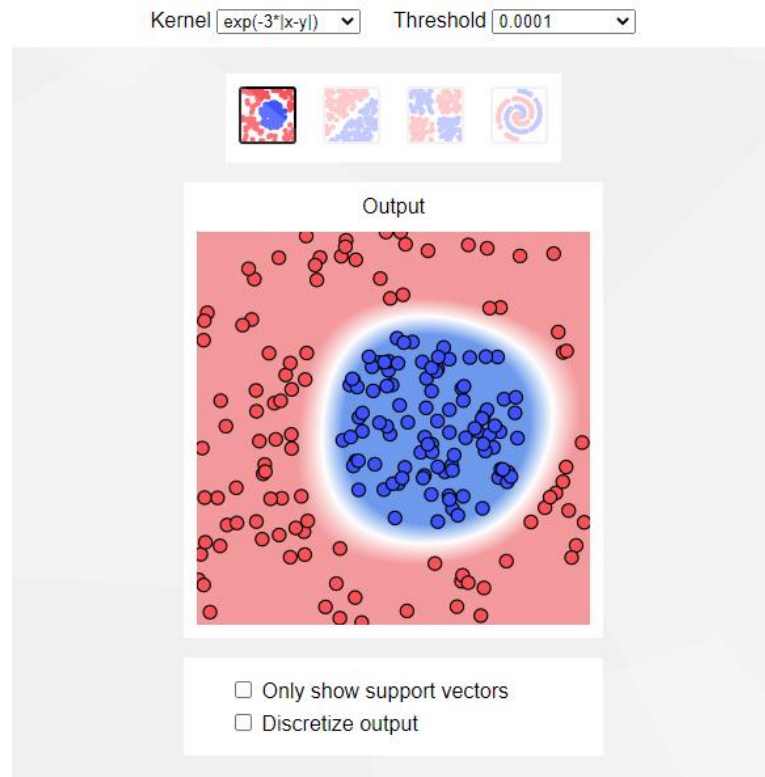
$$K(x, z) = \tanh(ax^T z + b)$$

Kernel



SVM Playground

❑ <http://macheads101.com/demos/svm-playground>



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